

Exact Entanglement-Breaking Time for a Thermal Qubit with Pure Dephasing: Flow, Choi Geometry, and Every Boundary

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Abstract

We solve the phase-covariant qubit GKSL semigroup with thermal excitation, relaxation, Hamiltonian phase, and pure dephasing on the entire nonnegative rate cone. With the convention that $(\gamma_\phi/2)(\sigma_z \rho \sigma_z - \rho)$ damps coherence at rate γ_ϕ , we give the exact density flow and Liouvillian spectrum. Its normalized Choi state is separable precisely when $p(1-p)(1-\eta)^2 \geq \eta^q$. This yields one finite entanglement-breaking time exactly for faithful thermal baths and an explicit radical when pure dephasing vanishes. We further prove the sharp trace-distance contraction coefficient and classify all one-sided, dephasing, unitary, and identity boundaries.

1 Frozen convention and exact semigroup

Use the ordered basis $|0\rangle, |1\rangle$, with $|0\rangle$ ground,

$$\sigma_z = |1\rangle\langle 1| - |0\rangle\langle 0|, \quad \sigma_- = |0\rangle\langle 1|, \quad \sigma_+ = |1\rangle\langle 0|.$$

For $\gamma_\downarrow, \gamma_\uparrow, \gamma_\phi \geq 0$ and $\omega \in \mathbb{R}$, consider

$$\mathcal{L}(\rho) = -i[(\omega/2)\sigma_z, \rho] + \gamma_\downarrow \mathcal{D}[\sigma_-]\rho + \gamma_\uparrow \mathcal{D}[\sigma_+]\rho + \frac{\gamma_\phi}{2}(\sigma_z \rho \sigma_z - \rho), \quad (1)$$

where $\mathcal{D}[L]\rho = L\rho L^\dagger - \frac{1}{2}\{L^\dagger L, \rho\}$. The factor $1/2$ in the last term is part of the model: pure dephasing alone sends ρ_{01} to $e^{(-\gamma_\phi + i\omega)t}\rho_{01}$. Set

$$\Gamma_1 = \gamma_\downarrow + \gamma_\uparrow, \quad \Gamma_2 = \frac{\Gamma_1}{2} + \gamma_\phi.$$

Theorem 1 (All-rate solution). *If $\Gamma_1 > 0$, put*

$$p = \frac{\gamma_\uparrow}{\Gamma_1}, \quad \eta = e^{-\Gamma_1 t}, \quad c = e^{(-\Gamma_2 + i\omega)t}.$$

Then, for every density matrix and $t \geq 0$,

$$\rho_{11}(t) = p + \eta(\rho_{11}(0) - p), \quad \rho_{01}(t) = c\rho_{01}(0). \quad (2)$$

The Liouvillian is diagonalizable on every parameter face and

$$\det(\lambda I - \mathcal{L}) = \lambda(\lambda + \Gamma_1)((\lambda + \Gamma_2)^2 + \omega^2). \quad (3)$$

Proof. In the matrix-unit coordinate order $(E_{00}, E_{01}, E_{10}, E_{11})$, direct substitution in (1) gives

$$[\mathcal{L}] = \begin{pmatrix} -\gamma_\uparrow & 0 & 0 & \gamma_\downarrow \\ 0 & -\Gamma_2 + i\omega & 0 & 0 \\ 0 & 0 & -\Gamma_2 - i\omega & 0 \\ \gamma_\uparrow & 0 & 0 & -\gamma_\downarrow \end{pmatrix}.$$

The population block and two scalar coherence blocks yield (2) and (3). They also supply independent eigenoperators even at accidental eigenvalue collisions. When $\Gamma_1 = 0$, nonnegativity forces $\gamma_\downarrow = \gamma_\uparrow = 0$, and the same block decomposition applies directly. $\square$

2 Choi geometry and the exact EB time

Let $|\Omega\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$ and use the normalized Choi state $J_t = (I \otimes \Phi_t)(|\Omega\rangle\langle\Omega|)$. From Theorem 1, in the order $|00\rangle, |01\rangle, |10\rangle, |11\rangle$,

$$J_t = \frac{1}{2} \begin{pmatrix} a & 0 & 0 & c \\ 0 & b & 0 & 0 \\ 0 & 0 & d & 0 \\ \bar{c} & 0 & 0 & e \end{pmatrix}, \quad (4)$$

where

$$a = 1 - p(1 - \eta), \quad b = p(1 - \eta), \quad d = (1 - p)(1 - \eta), \quad e = \eta + p(1 - \eta).$$

Define $q = 2\Gamma_2/\Gamma_1 \geq 1$, so $|c|^2 = \eta^q$.

Theorem 2 (Exact entanglement-breaking threshold). *For $\Gamma_1 > 0$, Φ_t is entanglement breaking if and only if*

$$p(1 - p)(1 - \eta)^2 \geq \eta^q. \quad (5)$$

If $0 < p < 1$, there is a unique $\eta_ \in (0, 1)$ solving equality, and*

$$t_{\text{EB}} = -\frac{\log \eta_*}{\Gamma_1}, \quad \Phi_t \text{ is EB} \iff t \geq t_{\text{EB}}.$$

If $\gamma_\phi = 0$, writing $r = p(1 - p)$ gives

$$\eta_* = \frac{1 + 2r - \sqrt{1 + 4r}}{2r}; \quad p = \frac{1}{2} : \eta_* = 3 - 2\sqrt{2}. \quad (6)$$

For $p = 0$ or $p = 1$, no finite t is EB, but the limiting constant channel is.

Proof. First, (4) is positive: all diagonal entries are nonnegative and

$$ae - |c|^2 = p(1 - p)(1 - \eta)^2 + \eta - \eta^q \geq 0.$$

Partial transpose moves c into the middle 2×2 block. Its determinant is

$$\frac{bd - |c|^2}{4} = \frac{p(1 - p)(1 - \eta)^2 - \eta^q}{4}.$$

For a two-qubit state, positivity under partial transpose is equivalent to separability; a channel is EB exactly when its Choi state is separable. This proves (5). For $r = p(1 - p) > 0$, $F(\eta) = r(1 - \eta)^2 - \eta^q$ obeys

$$F(0) = r > 0, \quad F(1) = -1, \quad F'(\eta) = -2r(1 - \eta) - q\eta^{q-1} < 0$$

on $(0, 1)$. Hence the root is unique; η decreases with time. Setting $q = 1$ and taking the root in $(0, 1)$ gives (6). At $p = 0, 1$, the left side of (5) is zero while $\eta^q > 0$ at finite time. $\square$

3 Sharp contraction and complete boundary atlas

For Hermitian traceless $\Delta = (v \cdot \sigma)/2$, $\|\Delta\|_1 = \|v\|_2$. The affine translation cancels on differences, while the Bloch linear part is a rotation composed with

$$\text{diag}(e^{-\Gamma_2 t}, e^{-\Gamma_2 t}, e^{-\Gamma_1 t}).$$

Its largest singular value is attained on a coordinate axis. Thus:

Proposition 1 (Sharp trace contraction). *For $\Gamma_1 > 0$,*

$$\sup_{\rho \neq \sigma} \frac{\|\Phi_t(\rho) - \Phi_t(\sigma)\|_1}{\|\rho - \sigma\|_1} = \max\{e^{-\Gamma_1 t}, e^{-\Gamma_2 t}\}.$$

It is strictly below one for $t > 0$. Hence $\text{diag}(1 - p, p)$ is the unique stationary state and every recurrent orbit is constant.

Proof. The singular-value calculation gives the upper bound and a sufficiently small pair of Bloch vectors separated along a maximizing axis gives equality. If a trajectory returned after $T > 0$, iterating the strict contraction at time T against the fixed state would force its initial distance to be zero. $\square$

The degeneracies are not obtained by dividing by Γ_1 :

Face	Fixed states	EB and recurrence
$\Gamma_1 > 0, 0 < p < 1$	unique faithful thermal state	one finite EB time; no nonconstant recurrence
$\Gamma_1 > 0, p = 0$ or 1	unique pure endpoint	EB only as $t \rightarrow \infty$; no nonconstant recurrence
$\Gamma_1 = 0, \gamma_\phi > 0$	all diagonal states	complete dephasing is EB only at infinity; no nonconstant recurrence
$\Gamma_1 = \gamma_\phi = 0, \omega \neq 0$	all diagonal states	unitary; coherent states have period $2\pi/ \omega $; never EB
$\Gamma_1 = \gamma_\phi = \omega = 0$	every state	identity; never EB

The Liouvillian kernel has dimension one in the first two rows, two in the third and fourth, and four at the identity corner. This last distinction is essential: the zero generator does not have merely the diagonal kernel.

Source owners

The generator form is owned by Gorini–Kossakowski–Sudarshan (DOI [10.1063/1.522979](https://doi.org/10.1063/1.522979)) and Lindblad (DOI [10.1007/BF01608499](https://doi.org/10.1007/BF01608499)). The complete-positivity matrix criterion is due to Choi (DOI [10.1016/0024-3795\(75\)90075-0](https://doi.org/10.1016/0024-3795(75)90075-0)); the EB characterization follows Horodecki–Shor–Ruskai (DOI [10.1142/S0129055X03001709](https://doi.org/10.1142/S0129055X03001709)) and the two-qubit PPT theorem follows Horodecki–Horodecki–Horodecki (DOI [10.1016/S0375-9601\(96\)00706-2](https://doi.org/10.1016/S0375-9601(96)00706-2)). No priority claim is made for those standard mechanisms.